

Part A (18 marks). Each chart is a violin plot (white dot = median) made in R with ggplot2. For every violin the skew is identified and the relation of the mean to the median is stated.

How to read the plots: A violin is thick where many observations lie and thin in a tail. The skew is named after the side of the long, thin tail. A right (positive) skew has its tail towards high values, so the mean is pulled above the median. A left (negative) skew has its tail towards low values, so the mean is pulled below the median. A violin that tapers evenly on both sides is symmetric, so the mean is about equal to the median.

Chart 1: Daily steps (thousand steps)

Violin	Skew (answer)	Mean vs Median (answer)
Group 3	Left skew	Mean < Median
Group 5	Symmetric	Mean $\approx$ Median
Group 4	Right skew	Mean > Median

Interpretation: Group 3: Left skew, so Mean < Median. The median dot sits high (about 15.5) in the wide part of the violin at 14 to 17, and a thin tail stretches down to about 5. The tail is on the lower side, so the mean is pulled below the median. Group 5: Symmetric, so Mean $\approx$ Median. The violin is a single smooth bulge with the median dot (about 10) near the centre of a range of about 4 to 16, and the upper and lower halves look alike. Group 4: Right skew, so Mean > Median. The bulk is low (wide near 2 to 5, median about 4.5) and a long tail runs up to about 13. The tail is on the higher side, so the mean is pulled above the median.

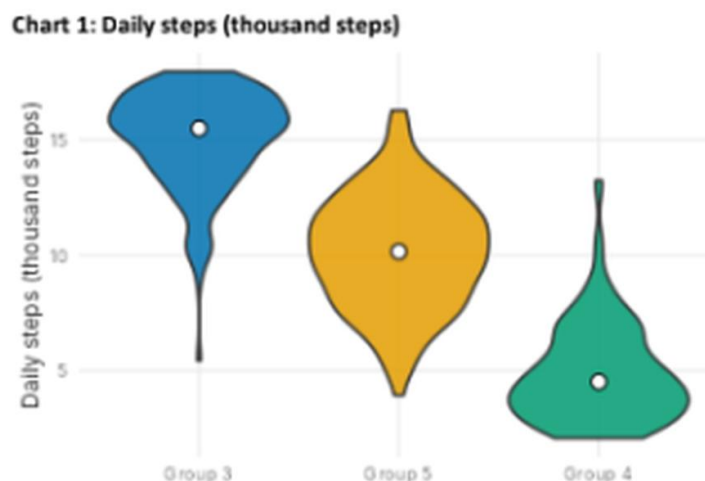


Chart 2: House price (Rs lakh)

Violin	Skew (answer)	Mean vs Median (answer)
Area 4	Right skew	Mean > Median
Area 2	Left skew	Mean < Median
Area 5	Symmetric	Mean $\approx$ Median

Interpretation: Area 4: Right skew, so Mean > Median. Most prices are bunched between about 20 and 60 (median about 47), with a long tail up to about 140. The tail is on the higher side, so the mean is above the median. Area 2: Left skew, so Mean < Median. Most prices are at the top end (wide near 170 to 200, median about 172), with a thin tail down to about 60. The tail is on the lower side, so the mean is below the median. Area 5: Symmetric, so Mean $\approx$ Median. The violin is a balanced bulge around the median (about 112), spanning about 45 to 185 with similar tapering above and below.



Chart 3: Delivery time (minutes)

Violin	Skew (answer)	Mean vs Median (answer)
Zone E	Symmetric	Mean $\approx$ Median
Zone S	Right skew	Mean $>$ Median
Zone W	Left skew	Mean $<$ Median

Interpretation: Zone E: Symmetric, so Mean $\approx$ Median. The violin is widest in the middle, with the median dot (about 49) near the centre of a range of about 20 to 80 and similar tapering on both sides. Zone S: Right skew, so Mean $>$ Median. Most deliveries are quick (wide near 10 to 25, median about 21) with a long thin tail to about 58. The tail is on the higher side, so the mean is above the median. Zone W: Left skew, so Mean $<$ Median. Most deliveries are slow (wide near 75 to 88, median about 78), with a tail down to about 47. The tail is on the lower side, so the mean is below the median.

Chart 3: Delivery time (minutes)

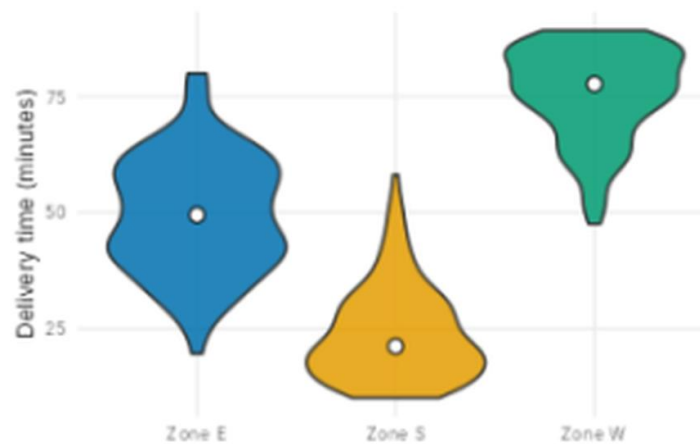


Chart 4: Order value (Rs hundred)

Violin	Skew (answer)	Mean vs Median (answer)
Store 3	Left skew	Mean < Median
Store 4	Right skew	Mean > Median
Store 5	Symmetric	Mean $\approx$ Median

Interpretation: Store 3: Left skew, so Mean < Median. The bulk is high (wide near 90 to 118, median about 100) with a tail running down to about 50. The tail is on the lower side, so the mean is below the median. Store 4: Right skew, so Mean > Median. The bulk is low (wide near 5 to 30, median about 25) with a tail up to about 73. The tail is on the higher side, so the mean is above the median. Store 5: Symmetric, so Mean $\approx$ Median. The violin is one balanced bulge around the median (about 62), from about 25 to 108, tapering evenly on both sides.



Part B (2 marks)

B1. In Chart 4, violin “Store 3” is not symmetric. Towards which side is its tail, and which is the better measure of a typical order value here: mean or median? Give one reason. [1]

Answer: The tail of Store 3 is towards the lower side (it is left-skewed): the violin is thick at about 90 to 118 and a thin tail runs down to about 50. The median (about 100) is the better measure of a typical order value, because the long low tail drags the mean below where most orders lie, so the mean understates the typical order, whereas the median is not affected by the size of the extreme values.

B2. Write one ggplot2 line that draws violins of value for each group from a data frame df (columns group, value), with the median shown. [1]

Answer:

```
ggplot(df, aes(x = group, y = value, fill = group)) + geom_violin() +  
stat_summary(fun = median, geom = "point", colour = "white", size = 3)
```

This maps group to the x-axis and value to the y-axis, draws one violin per group with `geom_violin()`, and `stat_summary(fun = median, geom = "point")` overlays the median of each group as a white dot, matching the white dot in the charts.

Final Answer Summary

Chart	Group	Skew	Mean vs Median
Chart 1	Group 3	Left skew	Mean < Median
Chart 1	Group 5	Symmetric	Mean $\approx$ Median
Chart 1	Group 4	Right skew	Mean > Median
Chart 2	Area 4	Right skew	Mean > Median
Chart 2	Area 2	Left skew	Mean < Median
Chart 2	Area 5	Symmetric	Mean $\approx$ Median
Chart 3	Zone E	Symmetric	Mean $\approx$ Median
Chart 3	Zone S	Right skew	Mean > Median
Chart 3	Zone W	Left skew	Mean < Median
Chart 4	Store 3	Left skew	Mean < Median
Chart 4	Store 4	Right skew	Mean > Median
Chart 4	Store 5	Symmetric	Mean $\approx$ Median